## Time Series Decomposition: Separating the Signal from the Noise

Time series decomposition breaks down a series into its fundamental components:

- **Trend**: The long-term progression
- **Seasonality**: Regular patterns that repeat at fixed intervals
- **Residual**: The remaining variation after removing trend and seasonality

This decomposition helps identify underlying patterns that might be hidden in the raw data:

```
from statsmodels.tsa.seasonal import seasonal_decompose

# Resample to monthly data for clearer decomposition
monthly_data = apple['Adj Close'].resample('M').last()

# Perform decomposition
decomposition = seasonal_decompose(monthly_data, model='multiplicative', period=12)

# Plot the decomposition
plt.figure(figsize=(14, 10))

plt.subplot(411)
plt.plot(monthly_data, label='Original')
plt.legend(loc='best')
plt.grid(True)

plt.subplot(412)
plt.plot(decomposition.trend, label='Trend')
plt.legend(loc='best')
plt.grid(True)

```